



# Geometrically Accurate Feed-Forward Gaussian Splatting for Unbounded Scenes

Master Thesis Defense

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*Autonomous Vision Group, University of Tübingen*

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**Prof. Dr. Gerard Pons-Moll**

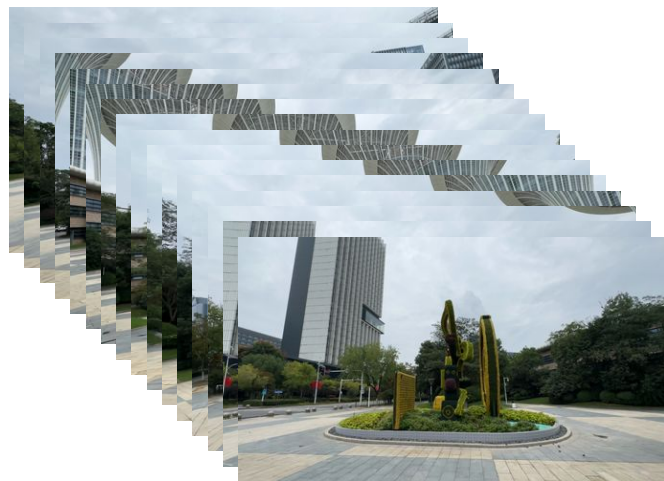
*Real Virtual Humans, University of Tübingen*

# 1

## Introduction

# Feed-Forward Gaussian Splatting

Feed-Forward Gaussian Splatting models take in multi-view images and output a Gaussian Splatting representation

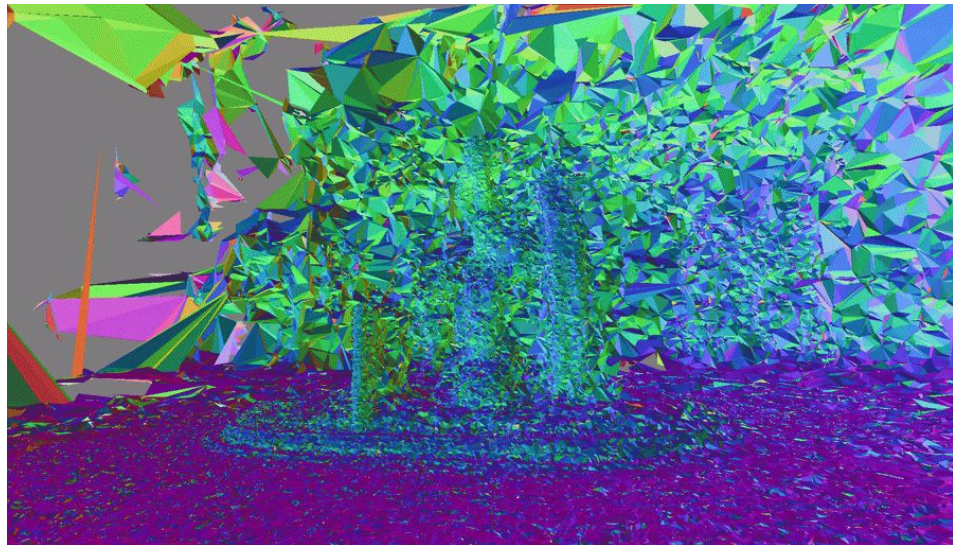


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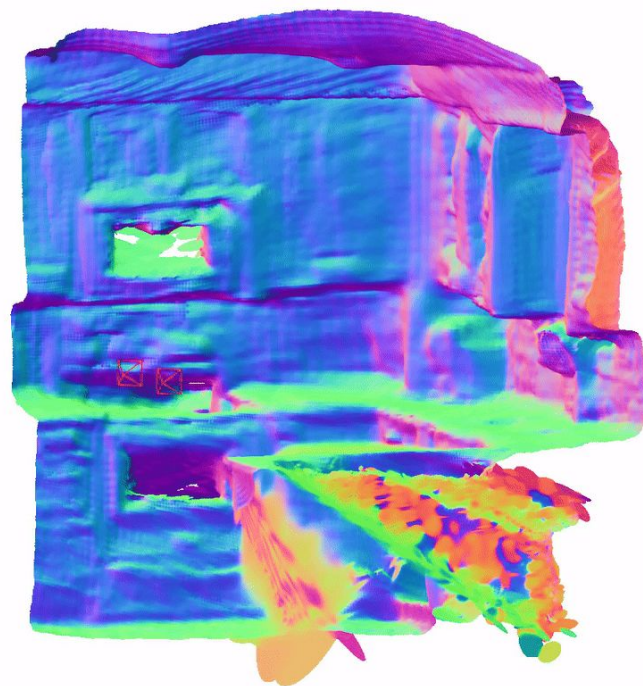
# Motivation

Feed-Forward Gaussian Splatting models achieve great visual quality but fail to achieve geometric quality, especially in the unbounded scene case





# Motivation



# Motivation



Goal: Enable Feed-Forward Gaussian Splatting models to reconstruct geometrically accurate 3D representations of unbounded scenes, as determined by the extracted mesh and rendered depth maps from the predicted Gaussians

# Index

## 1. Introduction

- **Data:** Unbounded scenes
- **Model**

## 2. Related Work

- Gaussian Splatting Renderers

## 3. Method

- Our additions to enable geometrically accurate 3D representations

## 4. Evaluation

- Metrics
- Extracted mesh and rendered depth maps
- Findings

# Datasets: Training

## DL3DV

- 10,510 multi-view scenes
- Around half of the scenes are unbounded
- Does not provide complete GT depth

## Depth Anything 3 massive dataset

- 160k environmental scenes
- 5% are outdoor
- 20% come from mixed datasets: indoor or outdoor
- GT depth comes from SfM methods thus incomplete



# Datasets: Training

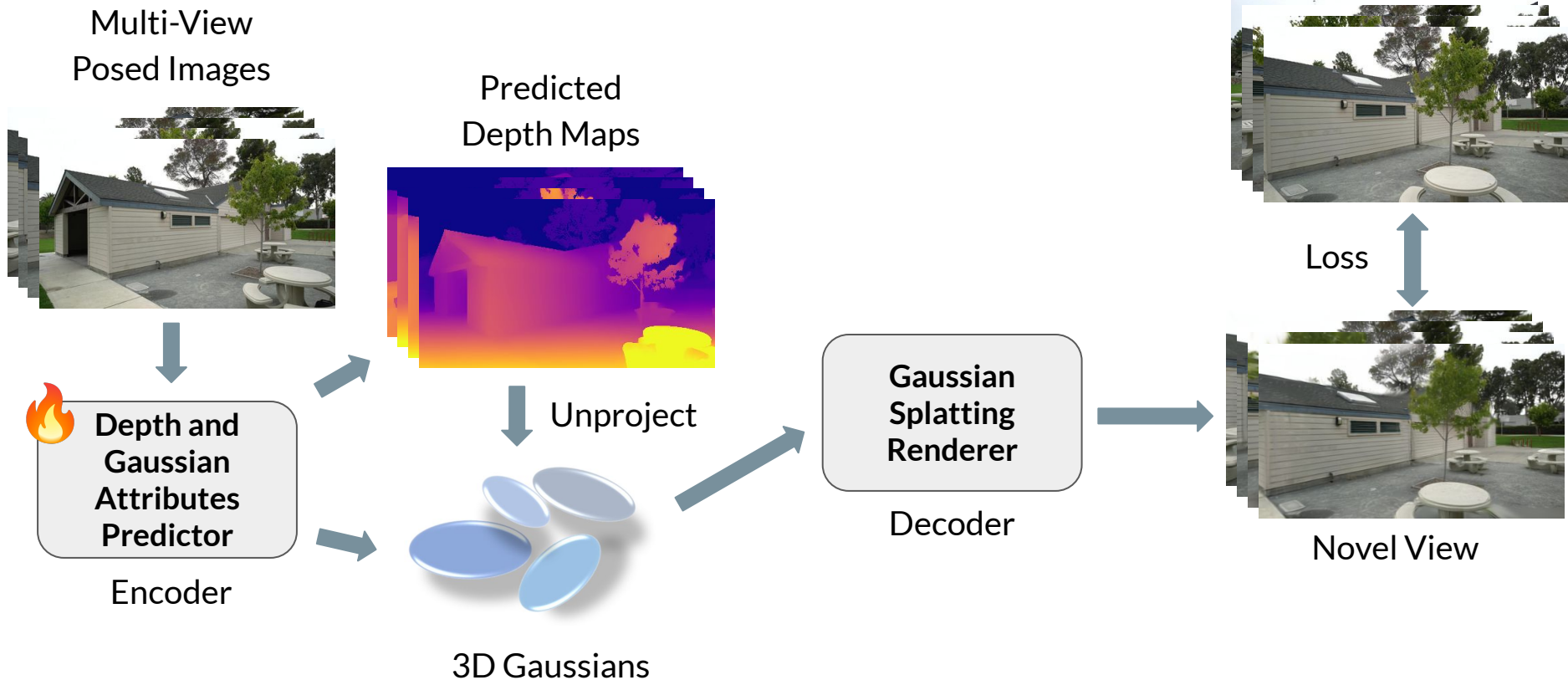
## DL3DV

- 10 510 multi-view scenes

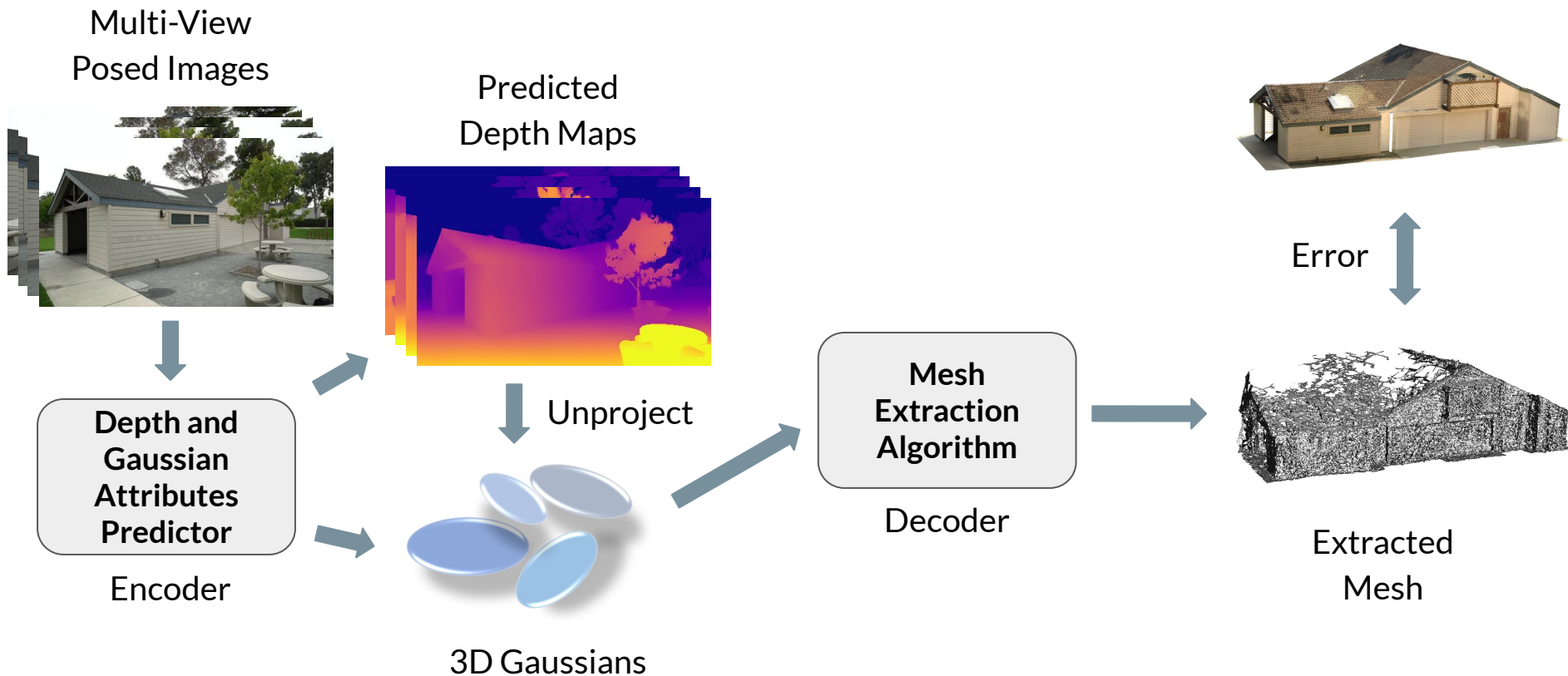
Main Hypothesis: Feed-Forward Gaussian Splatting models can achieve accurate 3D representations with photometric and self-supervised geometric losses

- 20% come from mixed datasets: indoor or outdoor
- GT depth comes from SfM methods thus incomplete

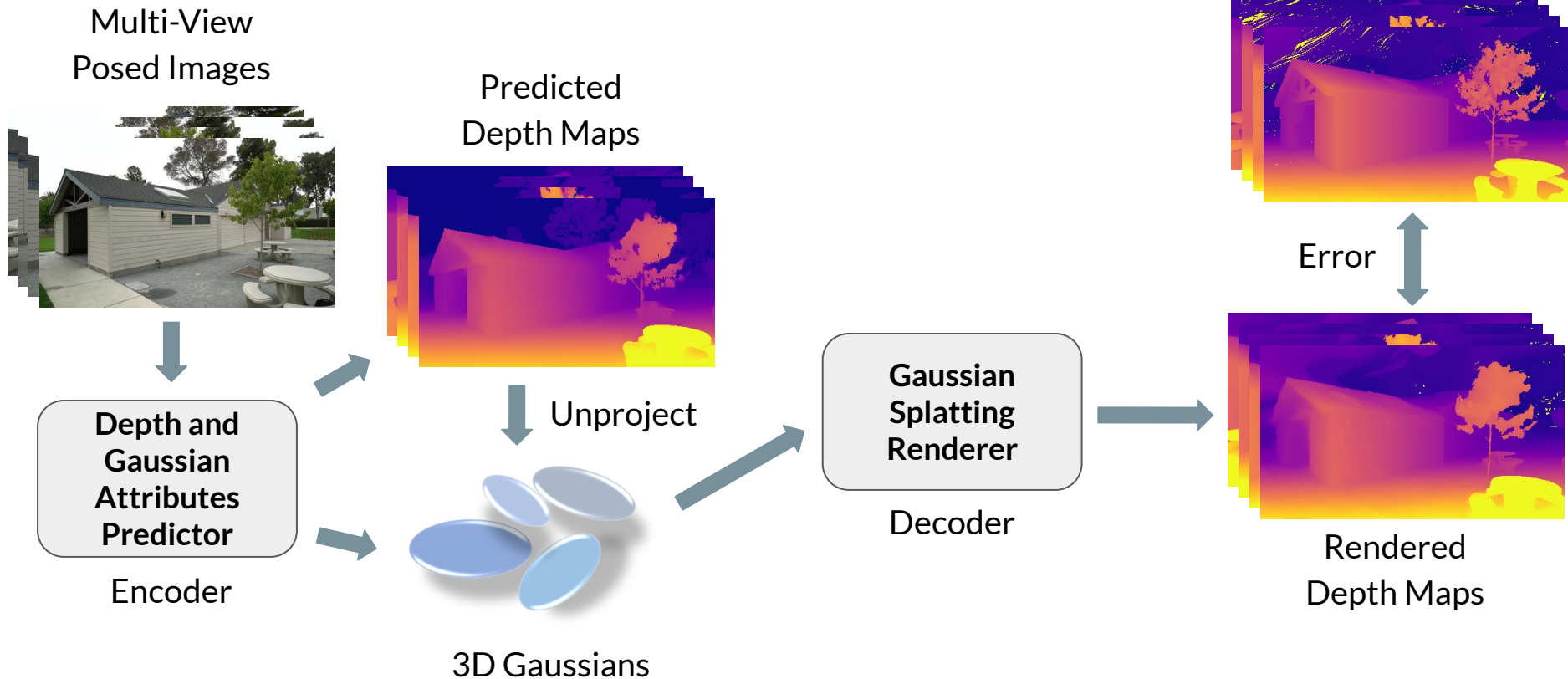
# Framework: Training with NVS loss



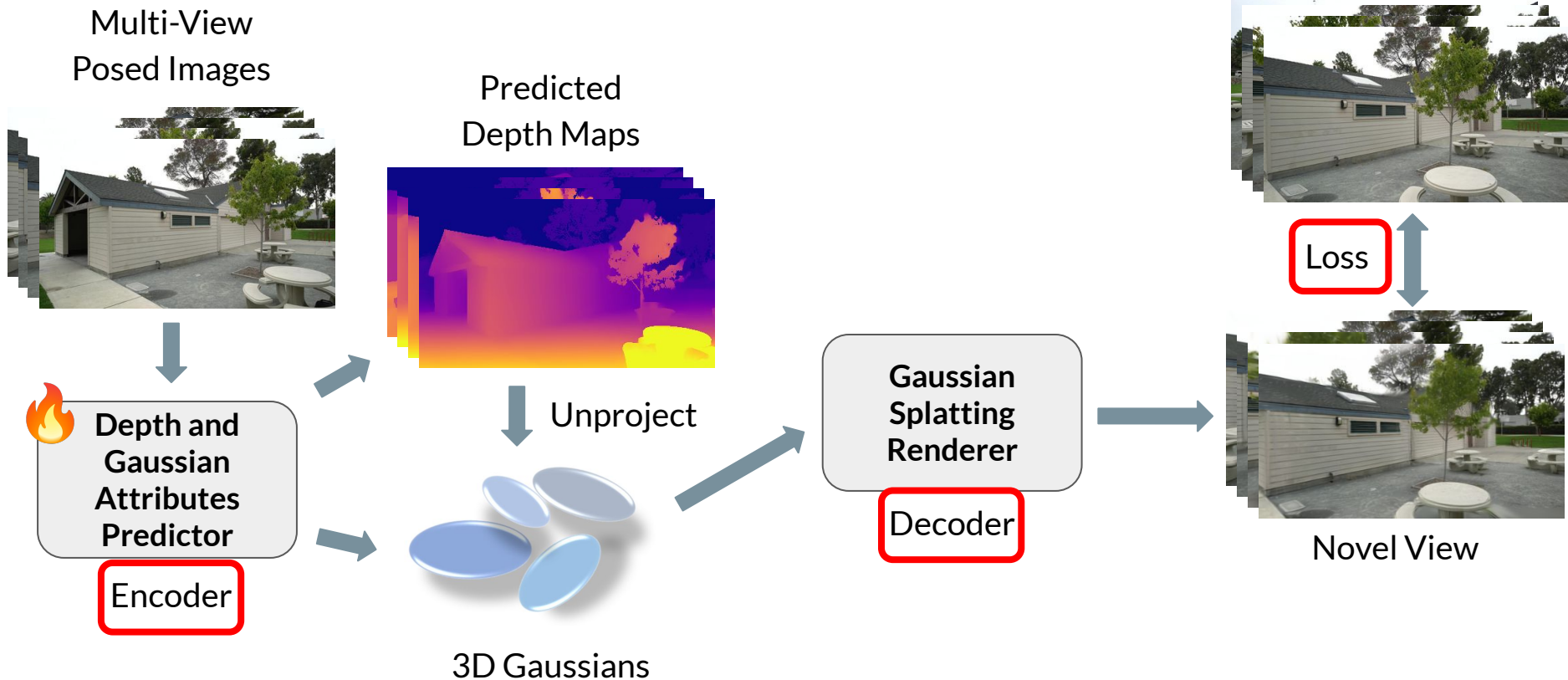
# Framework: Mesh Extraction



# Framework: Depth Estimation



# Framework: Training with NVS loss

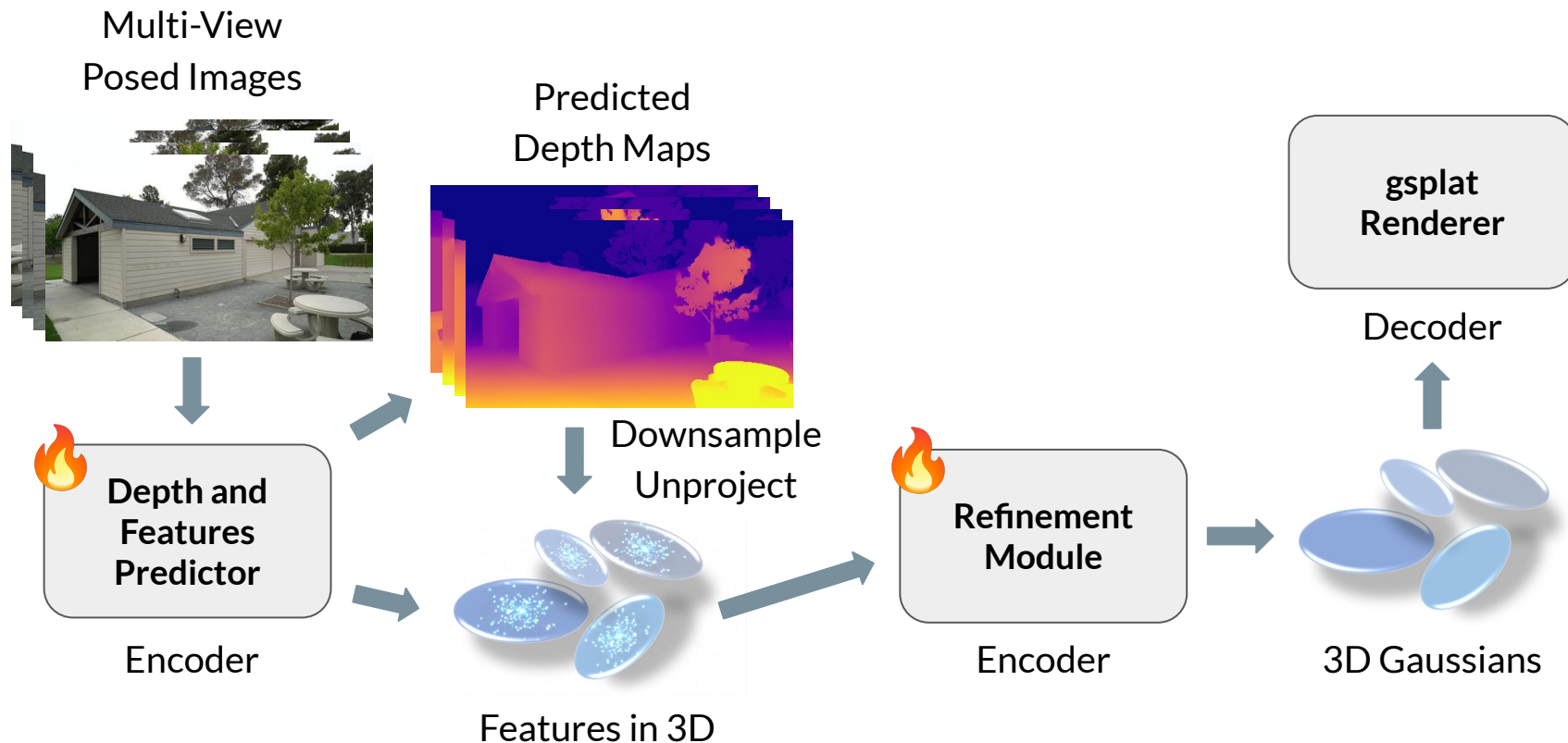


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Related Work



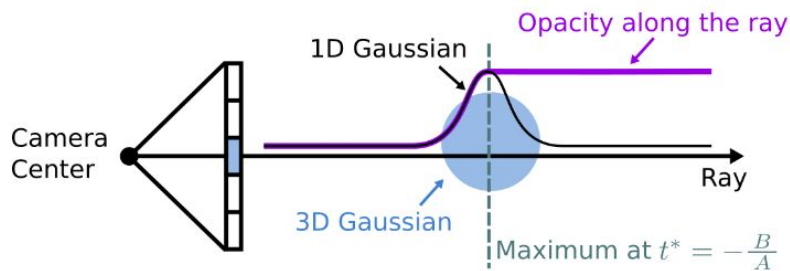
# ReSplat: Learning Recurrent Gaussian Splatting



# Gaussian Opacity Fields

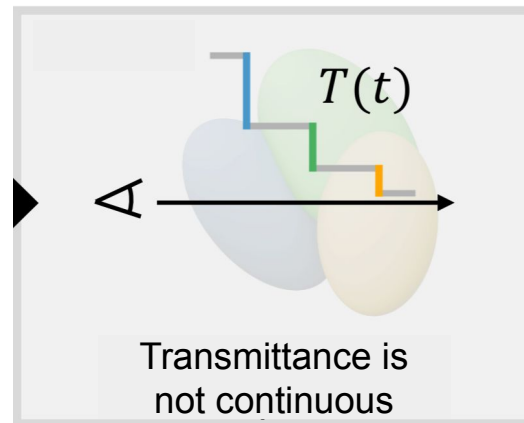
## Contributions

- Color, depth, and normal renderings
- Defines opacity at every 3D position enabling unbounded mesh extraction



## Shortcomings

- Depth rendering is not precise as it does not narrow down on 0.5-transmittance



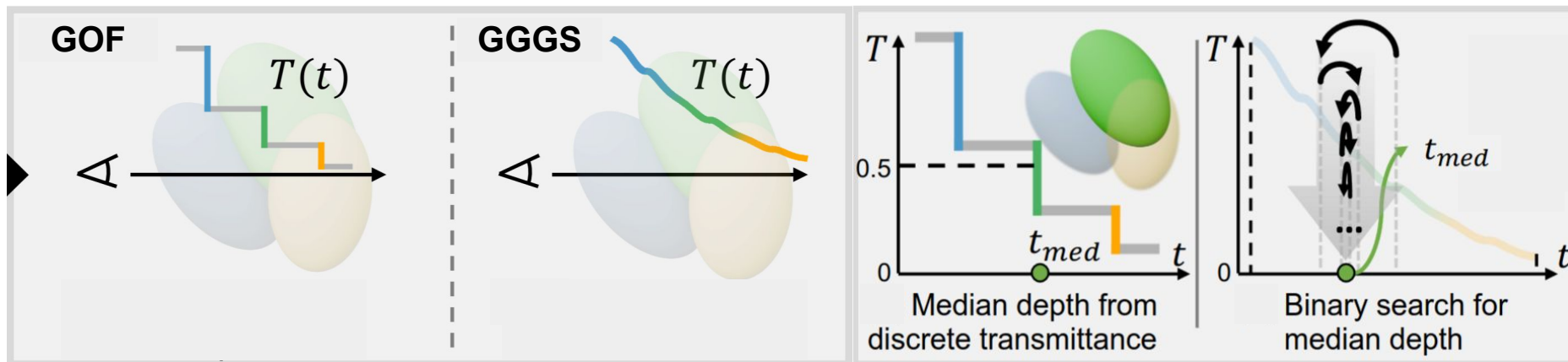
# Geometry-Grounded Gaussian Splatting

## Contributions

- Produces precise and consistent depth renderings

## Shortcomings

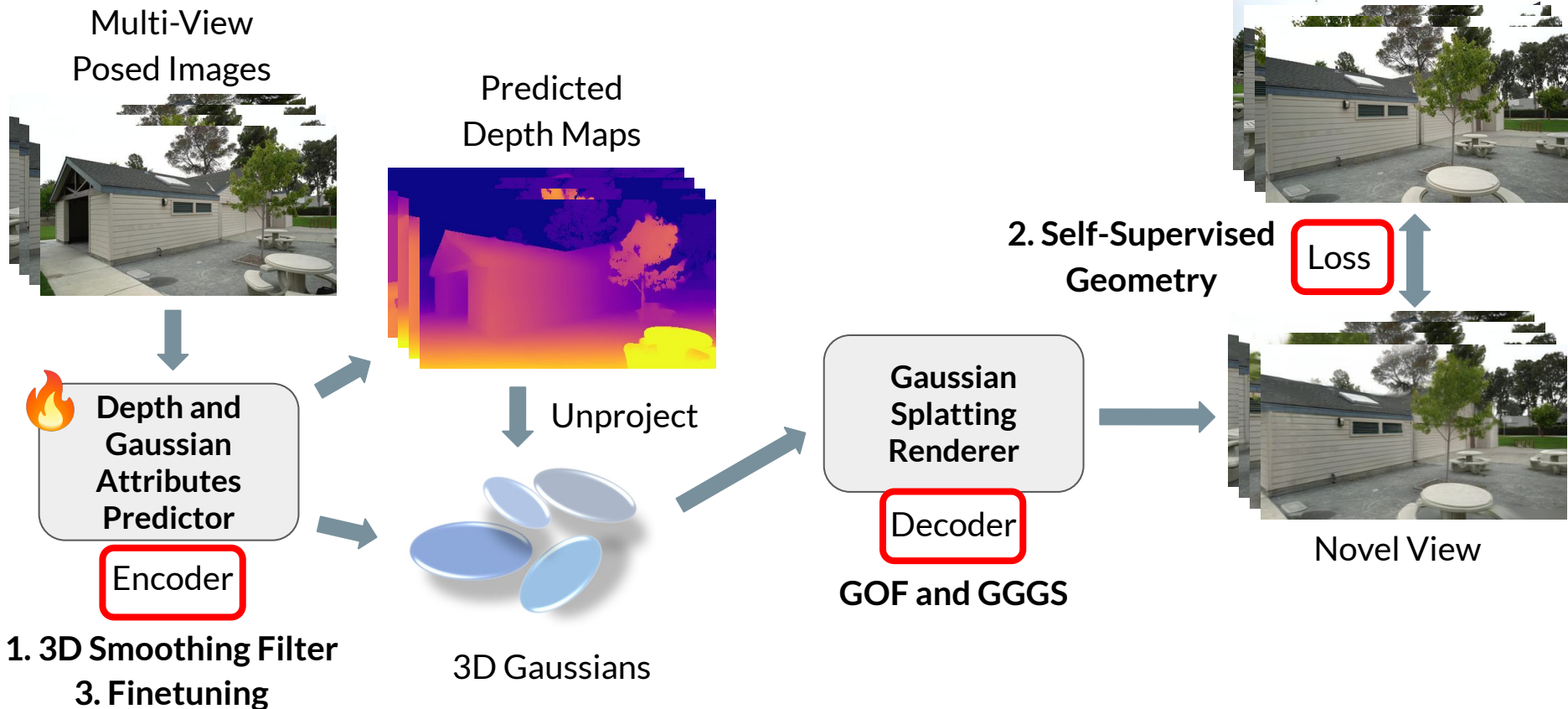
- Depth rendering process differs from color and normal rendering



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Method

# Framework: Additions



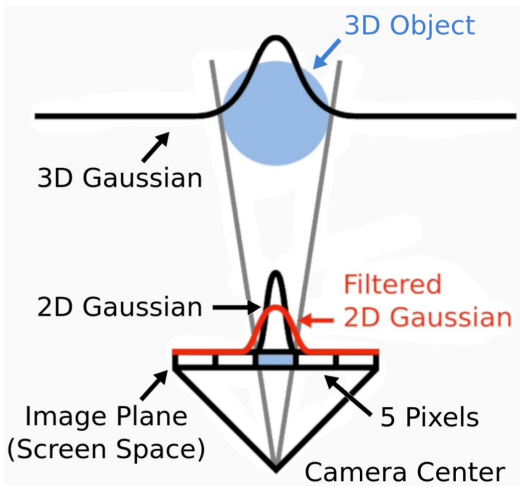
# 1. 3D Smoothing Filter



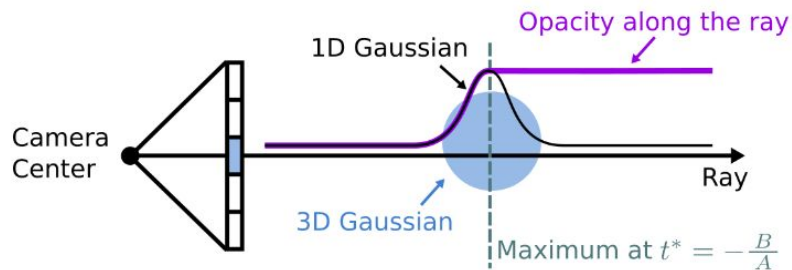
gsplat

GOF

# 1. 3D Smoothing Filter



gsplat



GOF



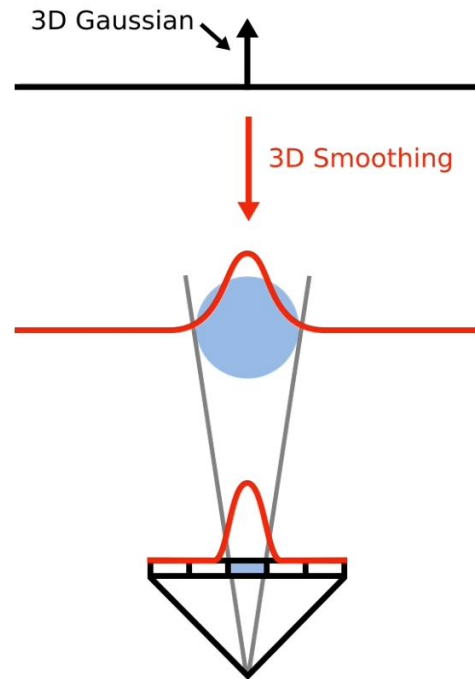
# 1. 3D Smoothing Filter

## MipSplatting on Feed-Forward Gaussian Splatting

- **Goal:** Eliminate high-frequency artifacts thus stabilizing training when using GOF
- **Issue:** 3D Gaussian frequency exceeds the camera's sampling rate
- **Solution:** Apply a 3D smoothing filter to cap the Gaussian's frequency below the maximal sampling limit

constrain  
frequency to  $\nu_k = \frac{f_n}{d_{k,n}}$

achieved by  $\Sigma_k = \Sigma_{k,\text{pred}} + \frac{s}{\nu_k} \cdot \mathbf{I}$



## 2. Self-Supervised Geometry Losses

### Normal consistency loss from 2DGS

- Ensures gaussians are locally aligned with surfaces
- Encourages alignment between gaussian normals and the gradient of the depth map

$$\mathcal{L}_n = \sum_i \omega_i (1 - \mathbf{n}_i^T \mathbf{N})$$

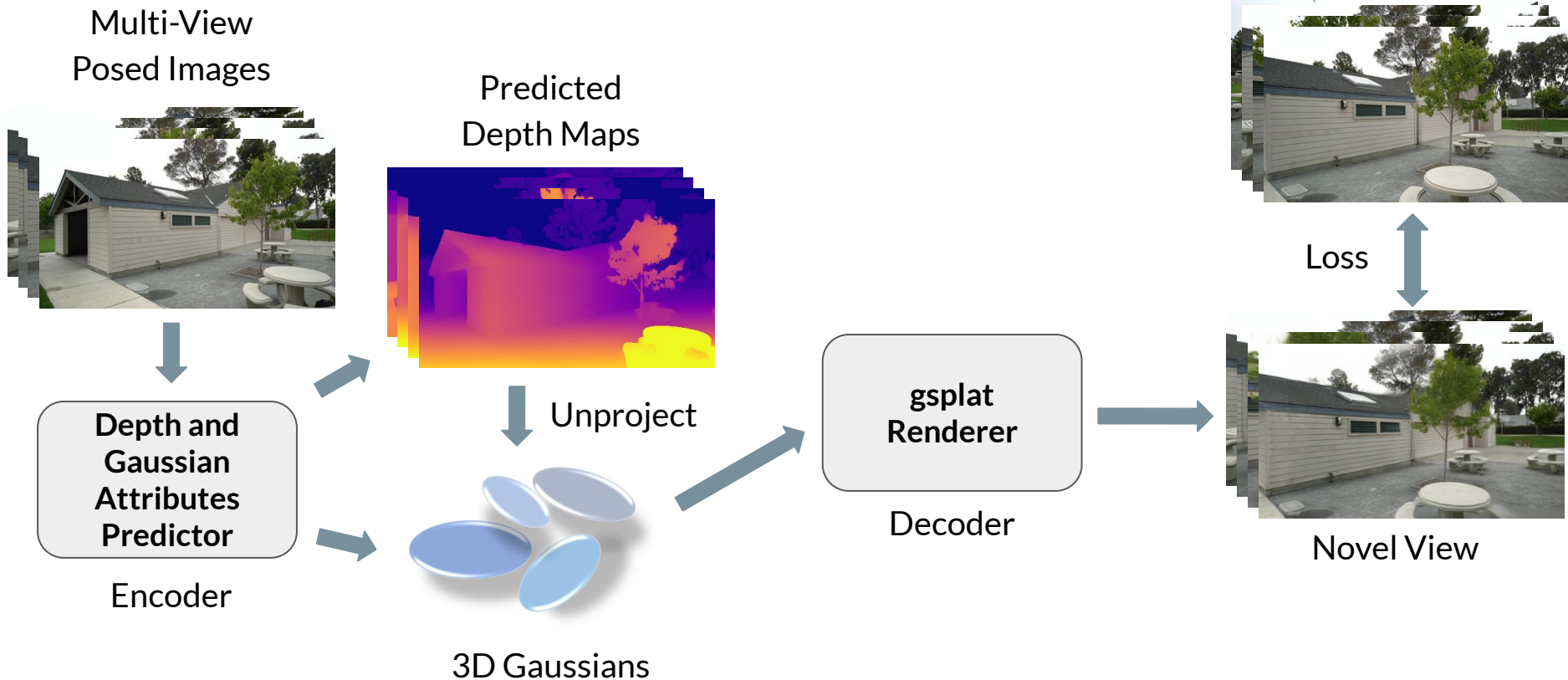
$$\mathbf{N}(x, y) = \frac{\nabla_x \mathbf{p}_s \times \nabla_y \mathbf{p}_s}{|\nabla_x \mathbf{p}_s \times \nabla_y \mathbf{p}_s|}$$

### Other losses are unstable and redundant

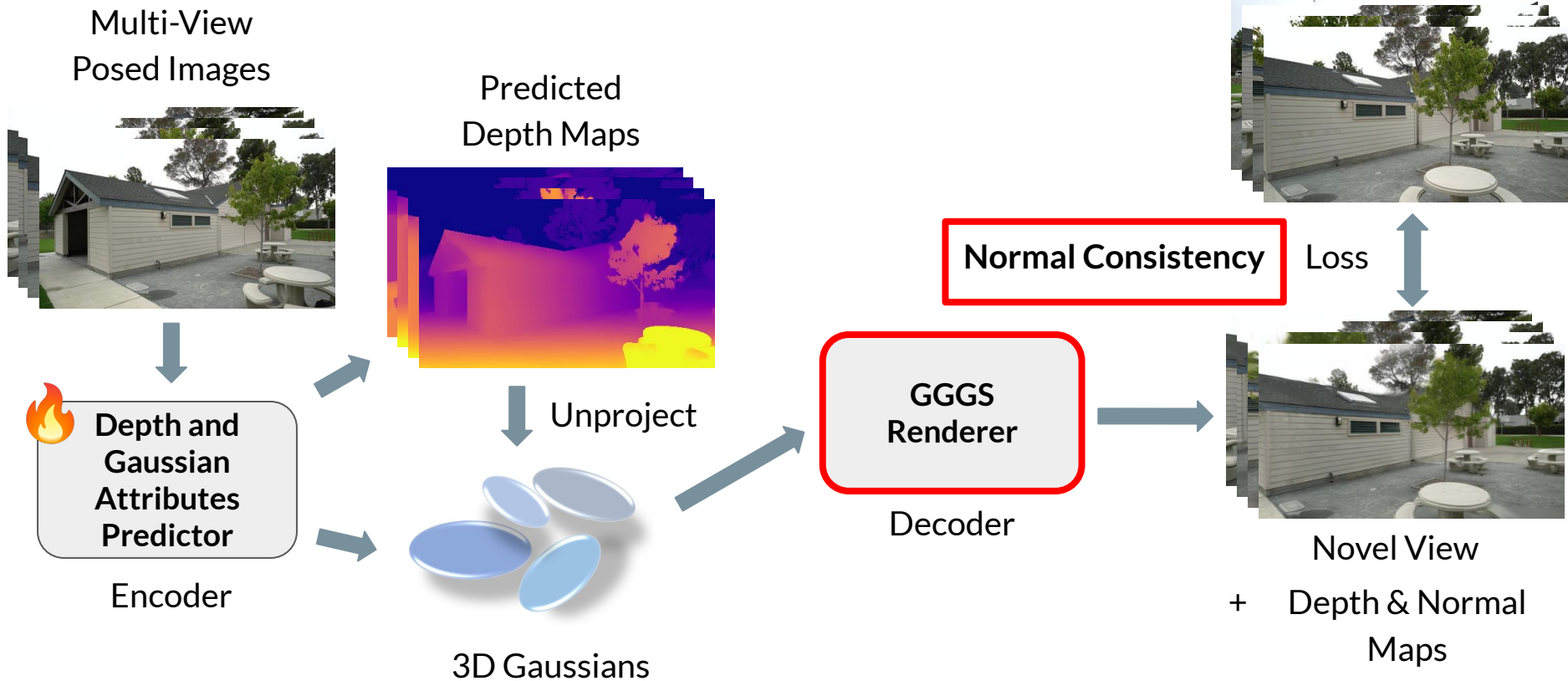
- Image Warp
- Edge-Aware Depth Smoothness

| GGGS Opt.          | w/o $\mathcal{L}_{iw}$ | w/o $\mathcal{L}_n$ | Full |
|--------------------|------------------------|---------------------|------|
| F-Score $\uparrow$ | 0.60                   | 0.57                | 0.60 |

### 3. ReSplat Encoder Fine-Tuning



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### 3. ReSplat Encoder Fine-Tuning

#### Forward pass: Color rendering in GGGS and gsplat

- Project 3D Gaussians to 2D view-space Gaussians
- Filter 2D covariances to avoid stability issues

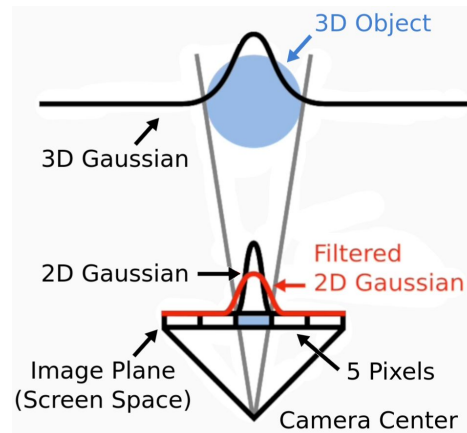
#### Backward pass: 2D Gaussian Filtering

##### GGGS

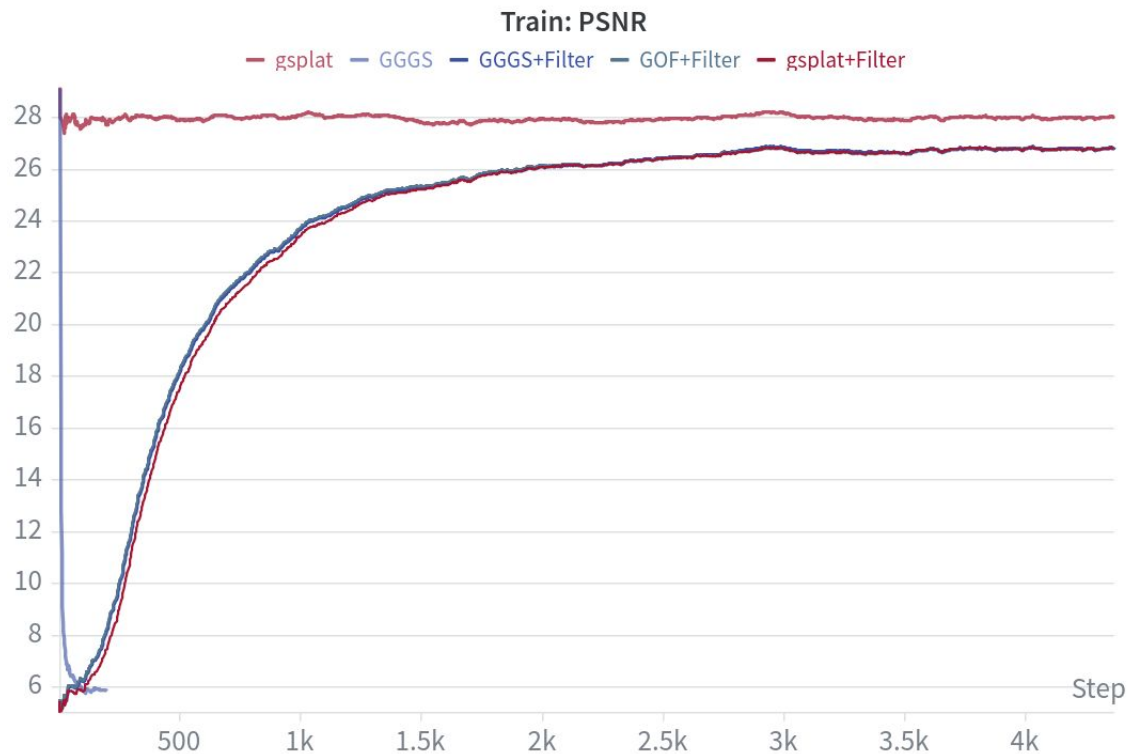
- Computes derivatives through intermediate variables
- Stops gradient flow when dividing by near-zero values

##### gsplat

- Simplifies chain rule math beforehand
- Avoids divisions which could lead to undefined or explosive gradients



### 3. ReSplat Encoder Fine-Tuning



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Evaluation

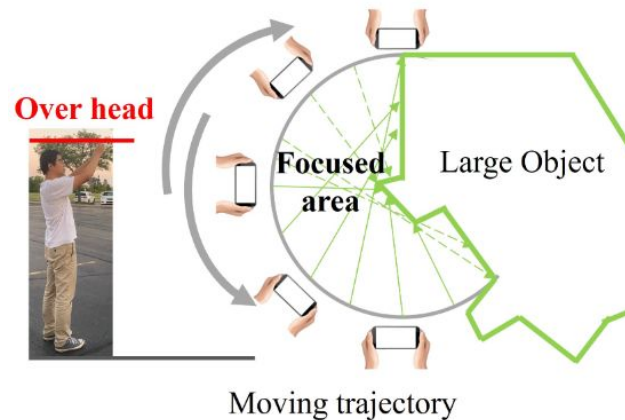


# Datasets: Testing

- **Tanks and Temples:** Barn, Caterpillar, Courthouse, Ignatius, Meetingroom, Truck
- **MipNeRF360:** bicycle, flowers, garden, stump, treehill
- **DL3DV's** test set: 10 outdoor scenes

## Context and target images

We select images which resemble the standardized camera trajectories of DL3DV



# Methods

- **Encoder Depth Prediction:** Trained on RE10K and DL3DV with photometric loss and fine-tuned on indoor and synthetic datasets with GT depth supervision
- **Encoder**
  - Trained with gsplat as decoder and photometric losses
  - Trained from scratch with GOF and normal consistency loss
  - Fine-tuned to GOF decoder and normal consistency loss
  - Fine-tuned to GGGS decoder and normal consistency loss
- **GGGS per-scene optimization for 4k steps and regularization from 1k steps**
  - With MV losses: Including image-warp like loss
  - Without MV losses: L1, SSIM, and normal consistency loss
- **SurfSplat**
- **Depth Anything 3 Giant**

# Metrics

## Mesh Reconstruction

Tanks and Temples provides GT point clouds and evaluation code. We report:

- F-Score: Percentage of points in reconstruction close enough to the ground truth and viceversa

## Depth Prediction

‘GT’ depth maps are rendered from the per-scene GGGS-optimized gaussian scene. We report:

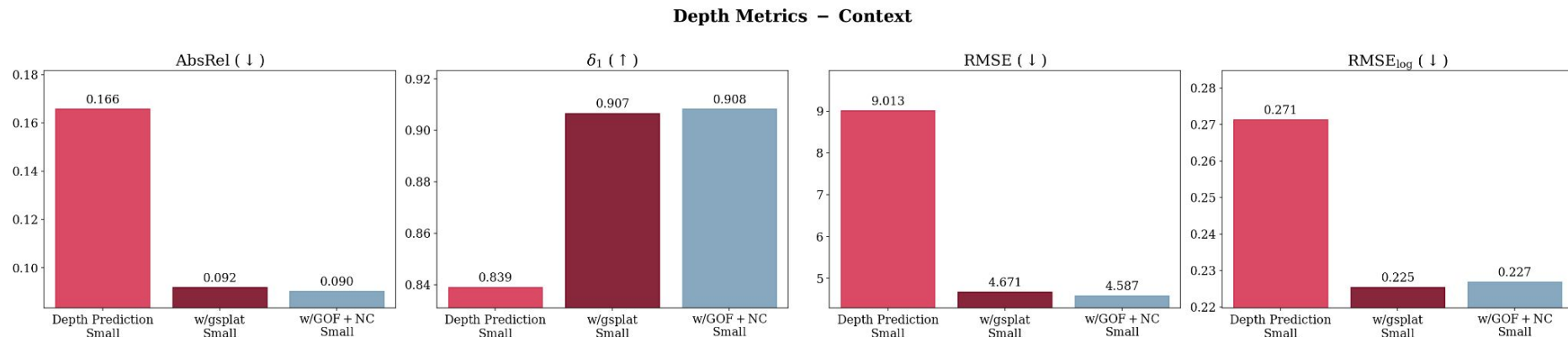
- Mean Absolute Relative L1 error
- RMSE and log RMSE

- Threshold Accuracy,  $\delta_1$

$$\max \left( \frac{d_{\text{gt}}}{d_{\text{rend}}}, \frac{d_{\text{rend}}}{d_{\text{gt}}} \right) < 1.25$$

# Finding 1:

## Encoder Trained with Photometric Loss Excels at Depth Accuracy



- Training with photometric and normal consistency losses using GOF decoder does not significantly improve depth accuracy

# Finding 1:

## Encoder Trained with Photometric Loss Excels at Depth Accuracy

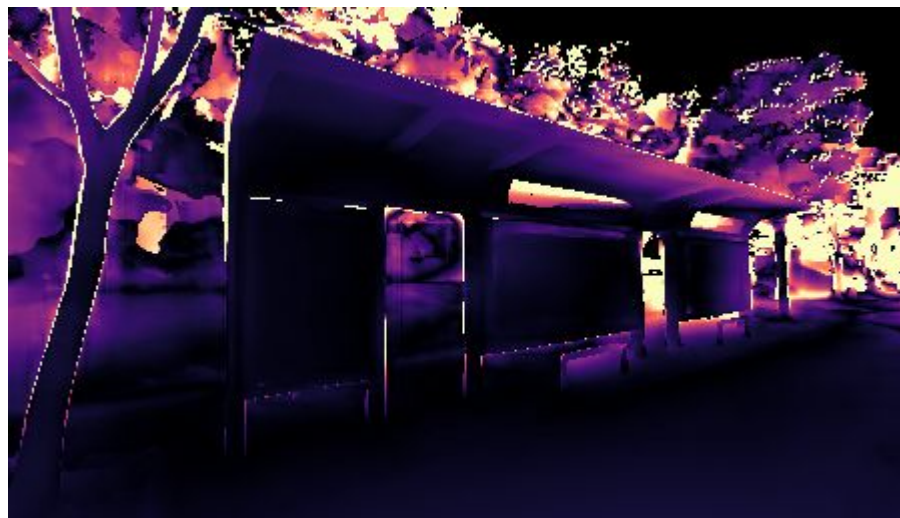
Input Images



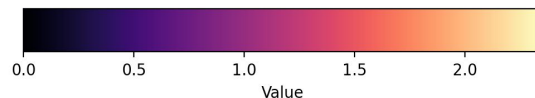
# Finding 1:

## Encoder Trained with Photometric Loss Excels at Depth Accuracy

Absolute Depth Difference, non-metric

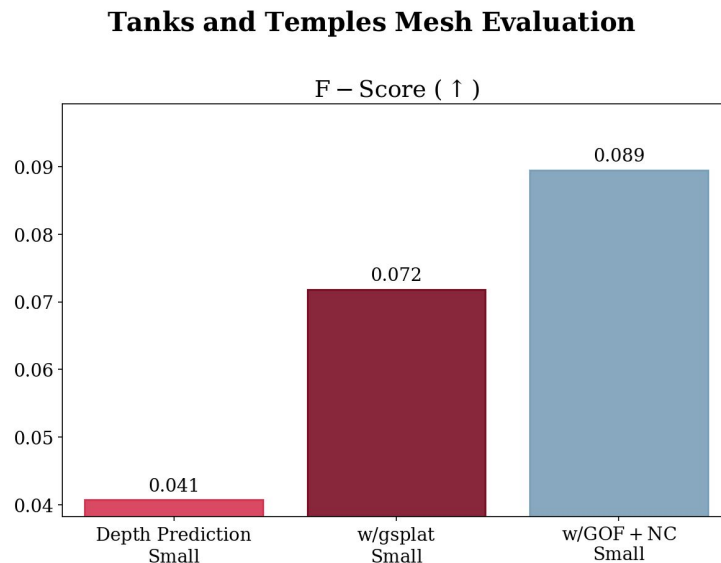


Scene AbsRel: 0.1829



Encoder w/gsplat 0.0612

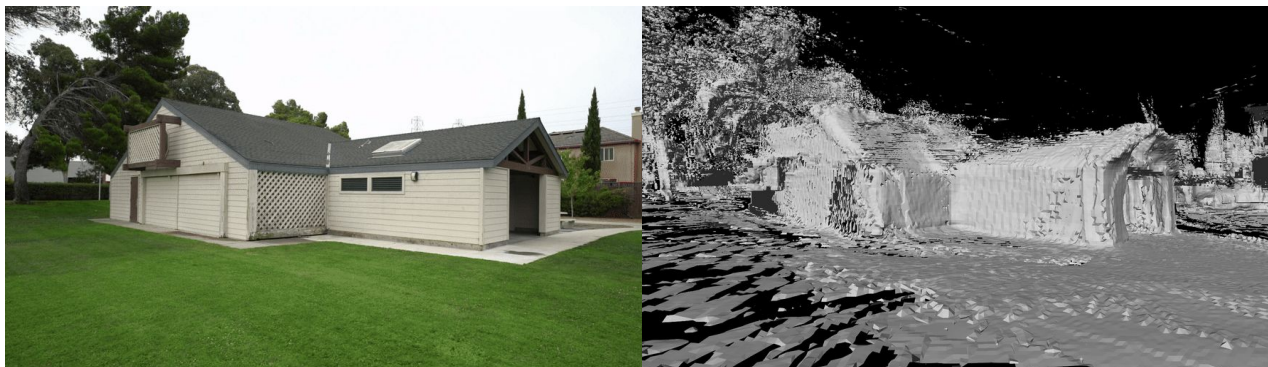
## Finding 2: Photometric and Normal Consistency Loss Enable Accurate Mesh Reconstruction





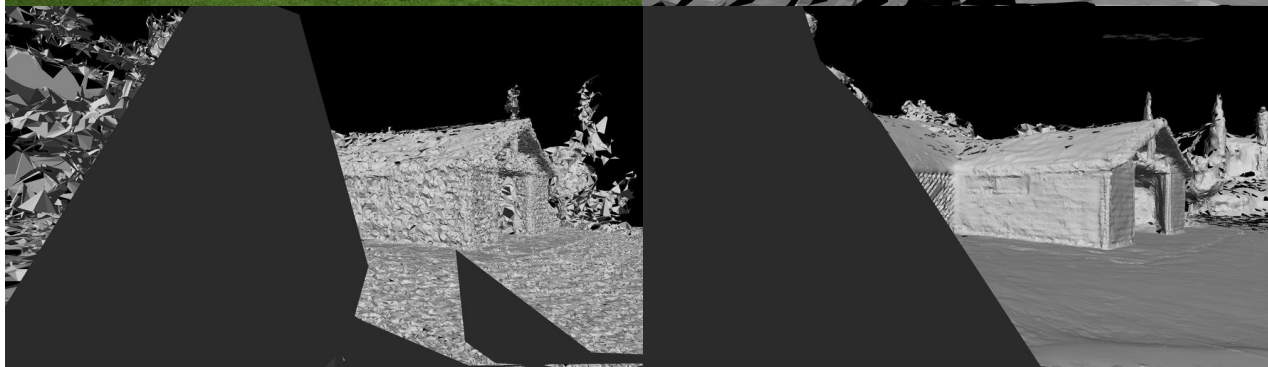
## Finding 2: Photometric and Normal Consistency Loss Enable Accurate Mesh Reconstruction

Color



Depth  
Prediction  
F-Score:  
0.0335

w/gsplat  
Small  
F-Score:  
0.0682

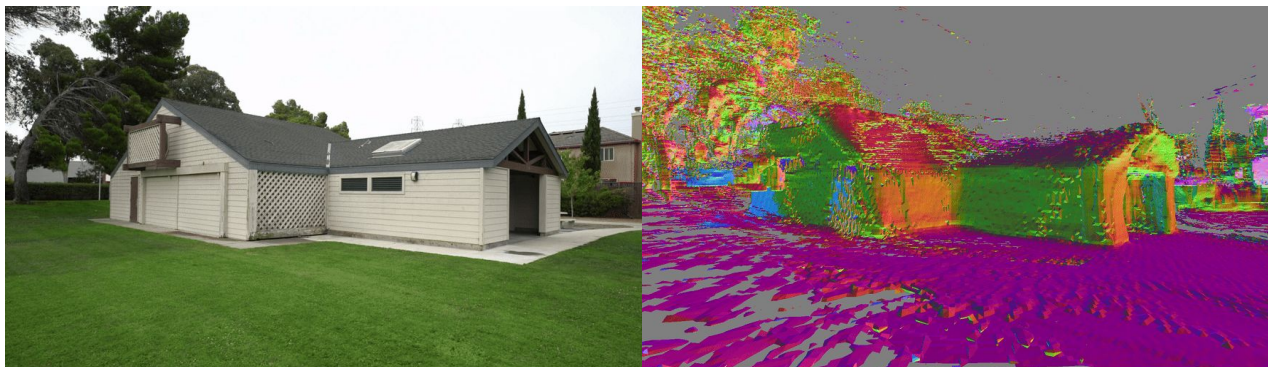


w/GOF  
Small  
F-Score:  
0.0738



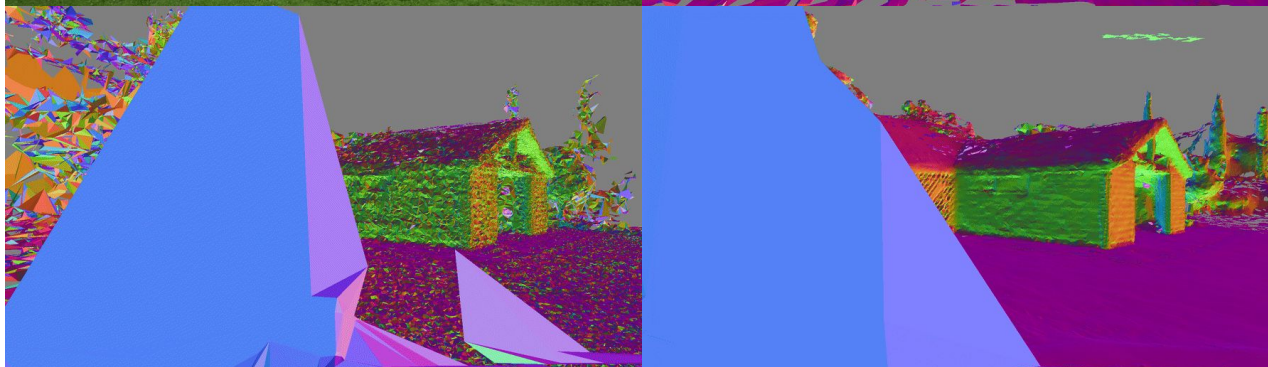
## Finding 2: Photometric and Normal Consistency Loss Enable Accurate Mesh Reconstruction

Color



Depth  
Prediction  
F-Score:  
0.033

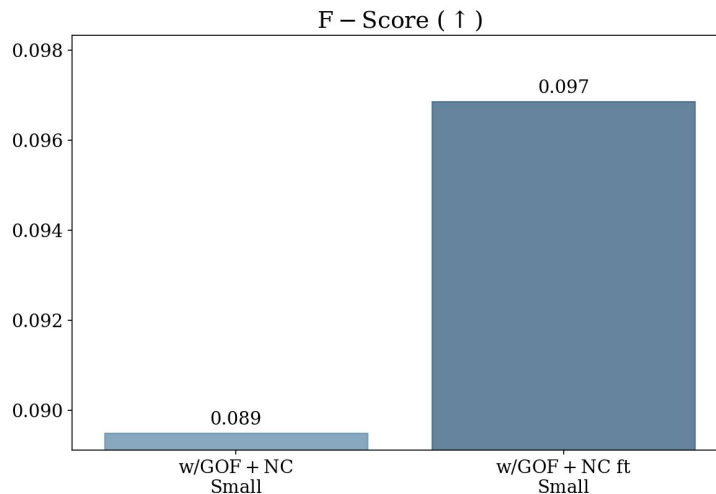
w/gsplat  
Small  
F-Score:  
0.068



w/GOF  
Small  
F-Score:  
0.073

# Finding 3: Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch

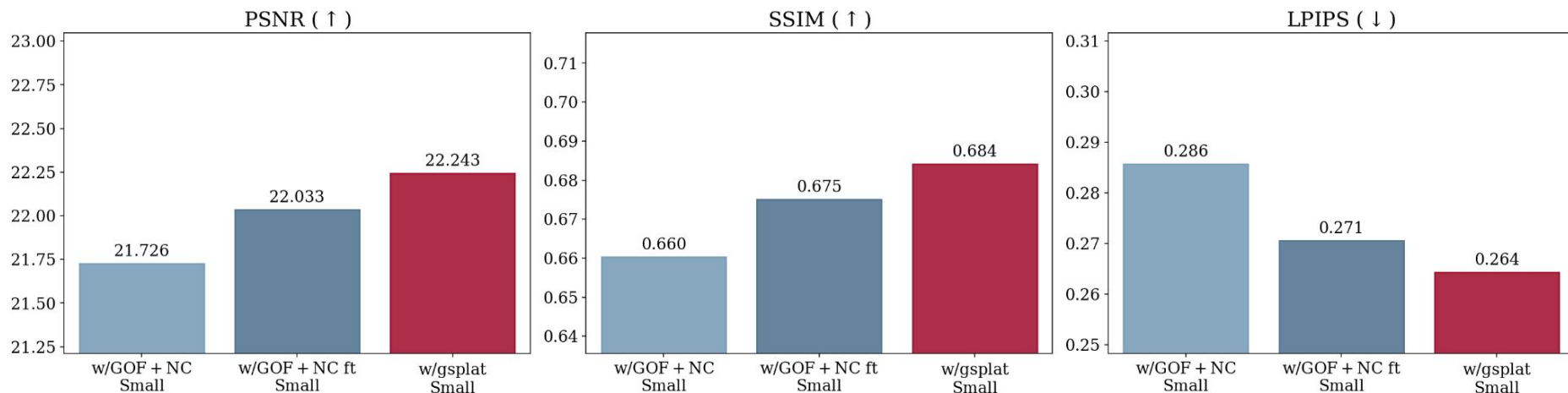
**Tanks and Temples Mesh Evaluation**



# Finding 3:

## Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch

NVS Metrics – Target



# Finding 3: Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch

GT



w/GOF+NC  
ft Small

Target views  
PSNR: 20.30



w/GOF+NC  
Small

Target views  
PSNR: 20.00

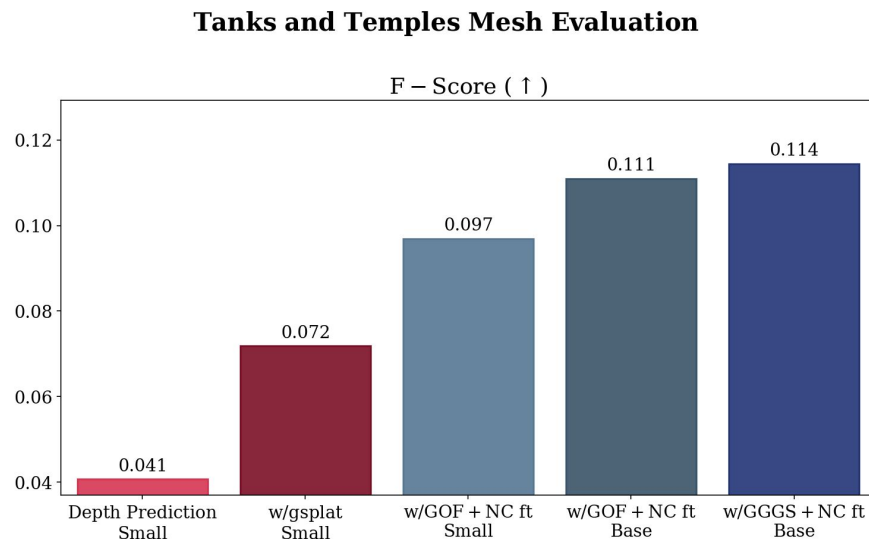


w/gsplat  
Small

Target views  
PSNR: 20.49



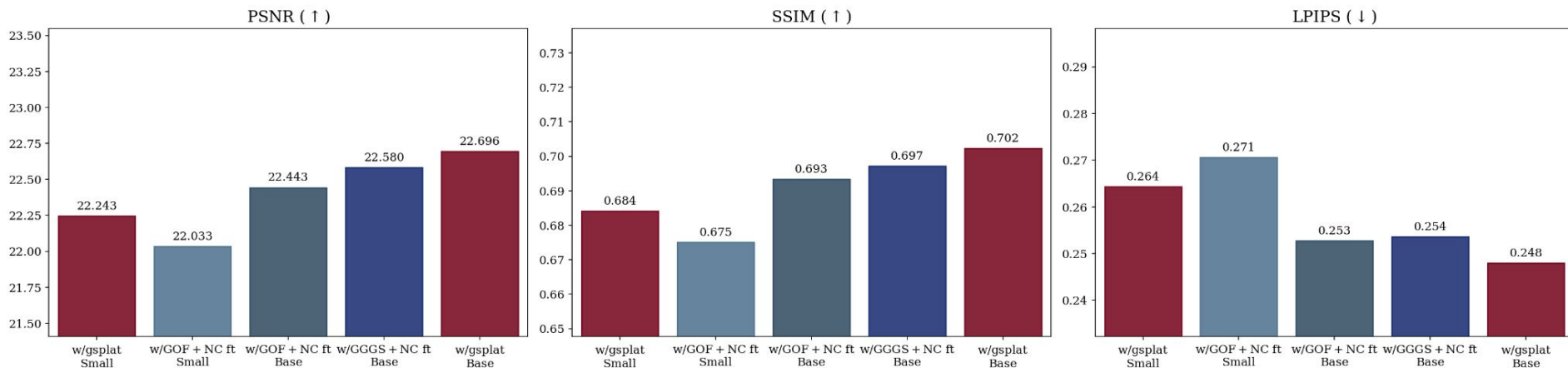
## Finding 4: Fine-Tuned Base Encoder with GGGS and Normal Consistency Loss Performs the Best



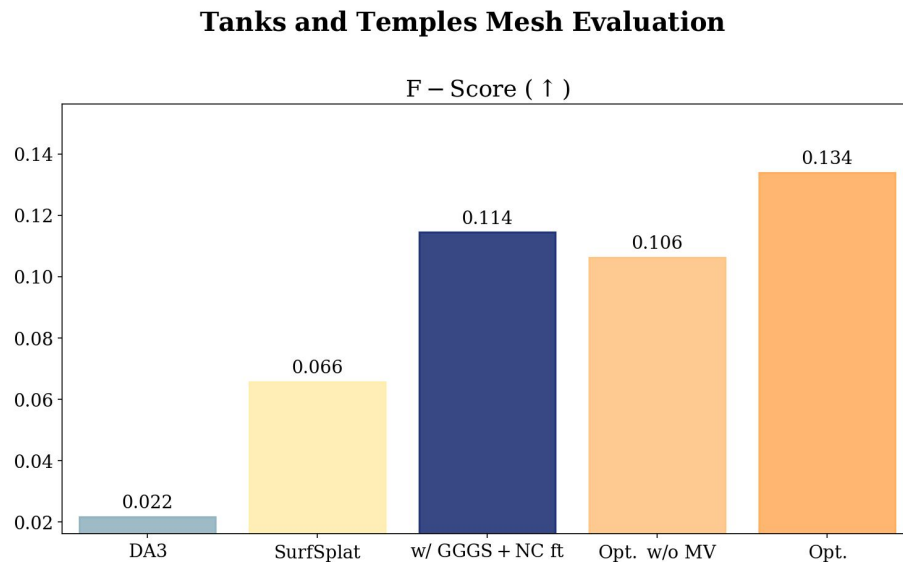
# Finding 4:

## Fine-Tuned Base Encoder with GGGS and Normal Consistency Loss Performs the Best

NVS Metrics – Target



## Finding 5: Fine-Tuned Base Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization



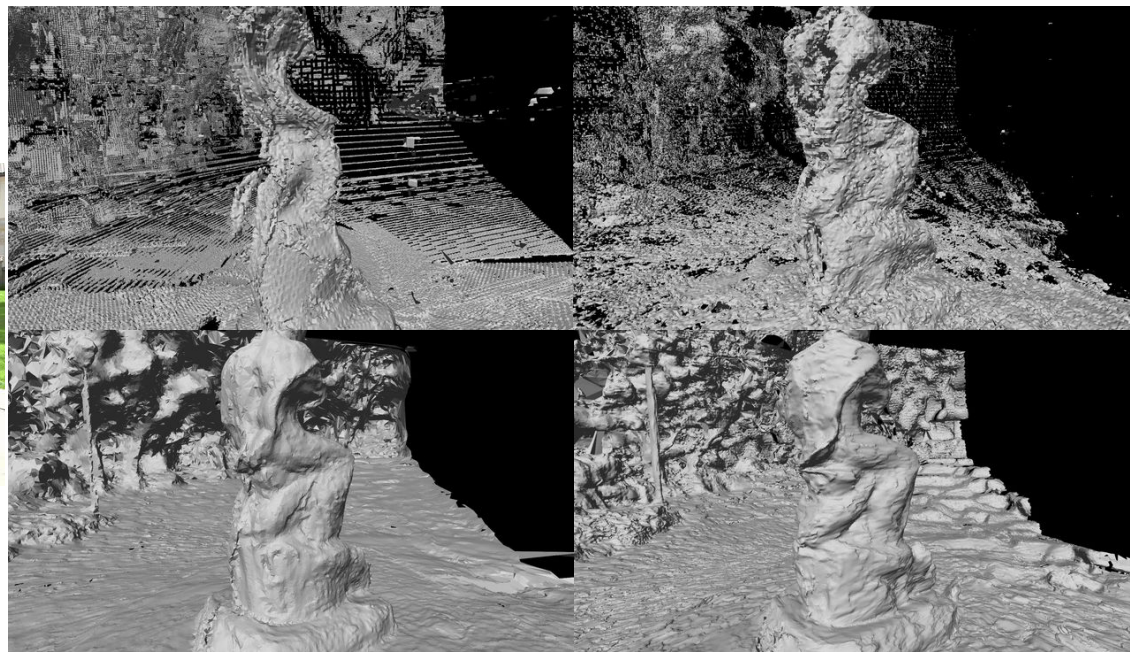


# Finding 5: Fine-Tuned ReSplat Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization

DA3. F-Score: 0.03

SurfSplat. F-Score: 0.12

Color

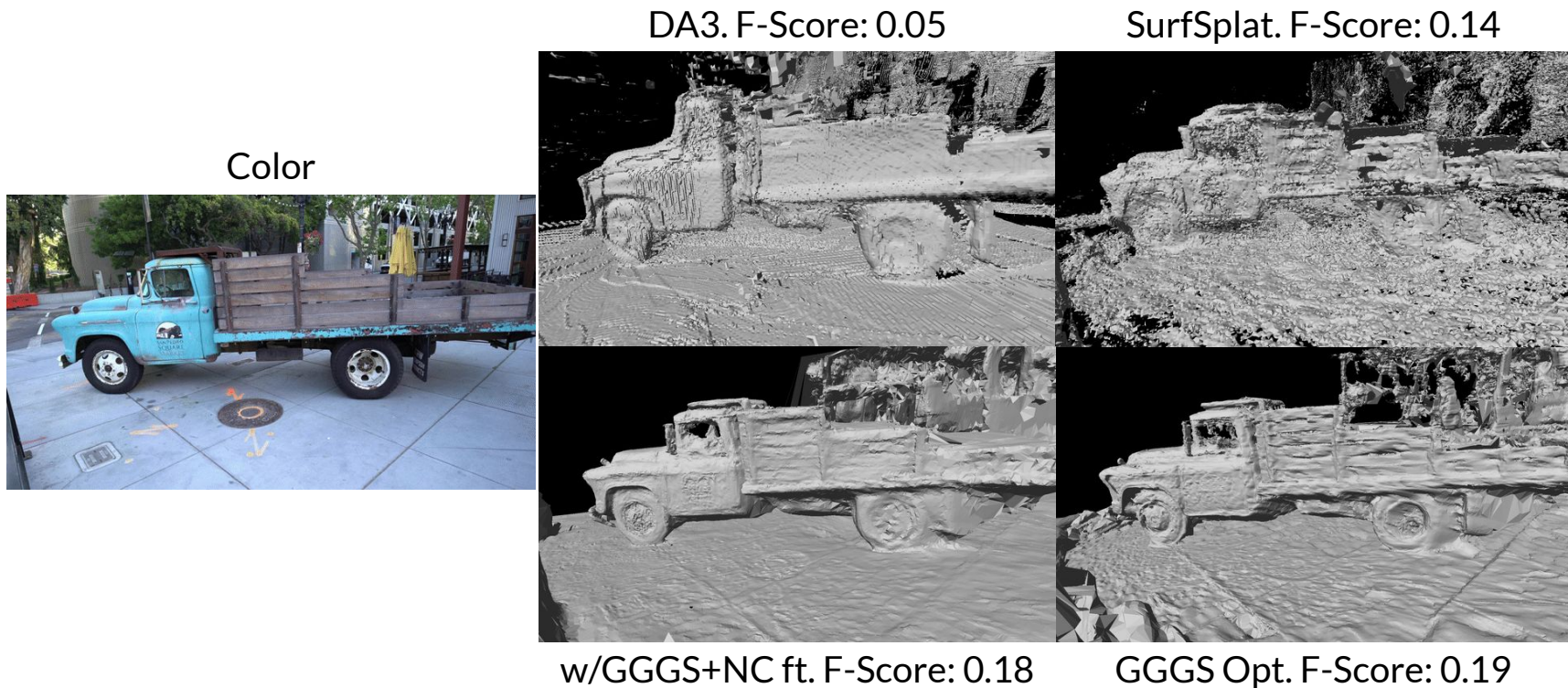


w/GGGS+NC ft. F-Score: 0.17

GGGS Opt. F-Score: 0.26

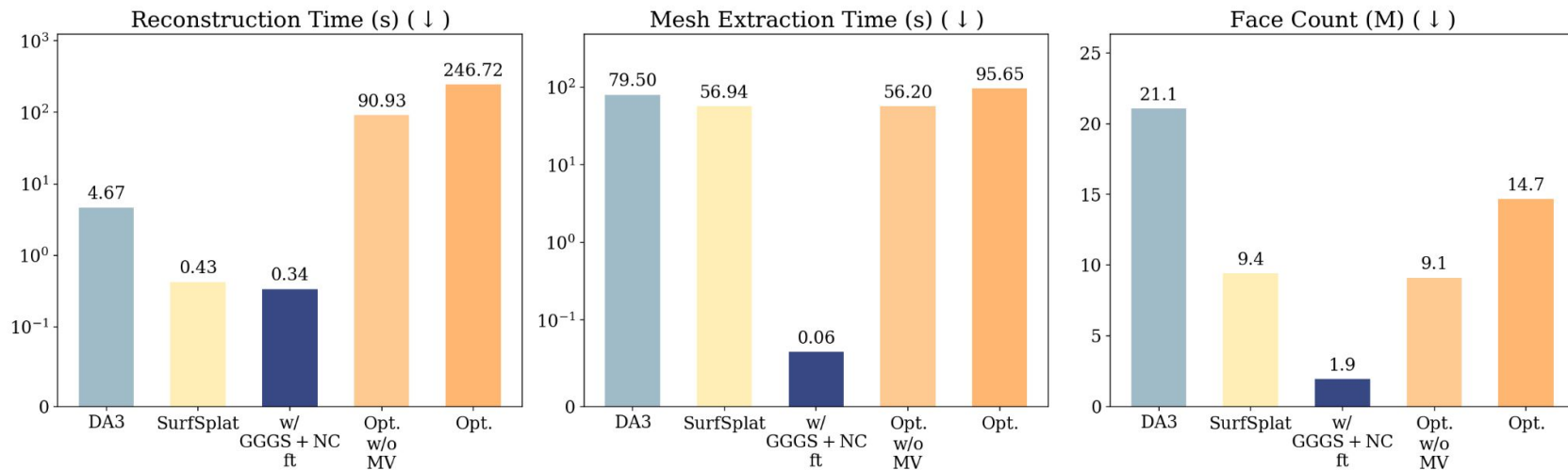


## Finding 5: Fine-Tuned ReSplat Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization



## Finding 5: Fine-Tuned ReSplat Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization

Metrics on Tanks and Temples



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Conclusion

# Conclusion

1. Feed-Forward Gaussian Splatting models **can** achieve accurate **depth estimation** with photometric losses
2. Normal consistency loss does not improve depth estimation significantly but does improve **mesh reconstruction accuracy**
3. Fine-Tuning Feed-Forward Gaussian Splatting models is not straightforward and depends on the choice of **primitives** and **renderers**
4. Feed-Forward Gaussian Splatting models **achieve state-of-the-art performance in depth estimation and mesh reconstruction tasks**

## Future Work:

- Compact 3D representations circumvent depth prediction
- Primitives other than gaussians: triangles or meshes
- New depth priors: Depth Anything V3



Fin

Questions ?



# Appendix

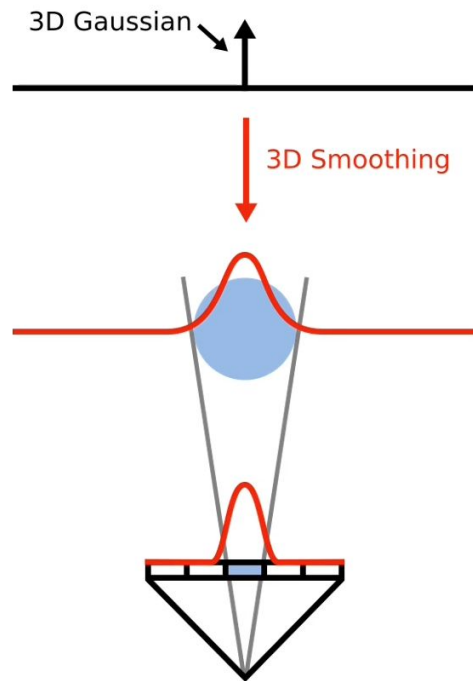
# 1. 3D Smoothing Filter

## MipSplatting on 3D Gaussian Splatting

- **Goal:** Eliminate high-frequency artifacts
- **Issue:** 3D Gaussian frequency exceeds the camera's sampling rate
- **Solution:** Apply a 3D smoothing filter to cap the Gaussian's frequency below the maximal sampling limit

constrain  
frequency to

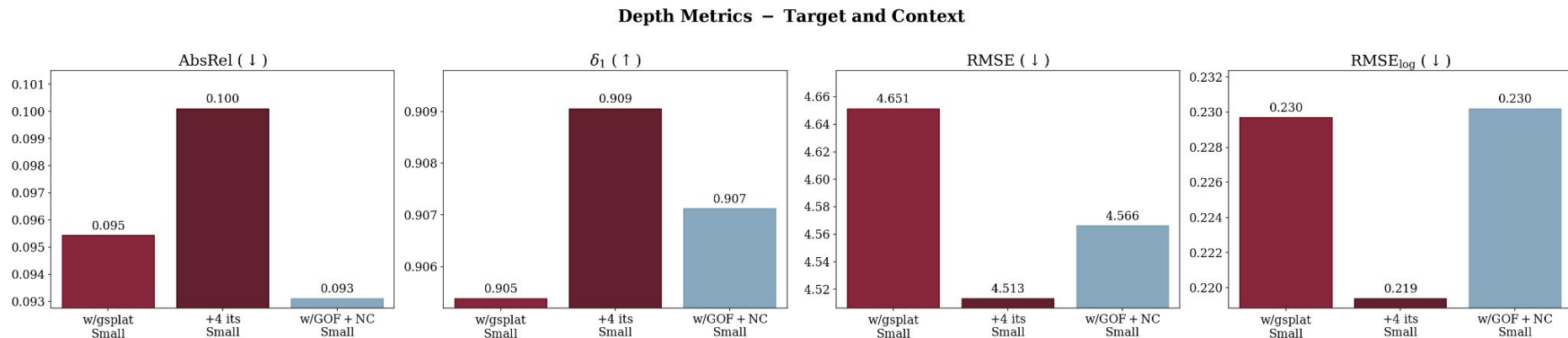
$$\hat{\nu}_k = \max \left( \left\{ \mathbb{1}_n(\mathbf{p}_k) \cdot \frac{f_n}{d_n} \right\}_{n=1}^N \right)$$





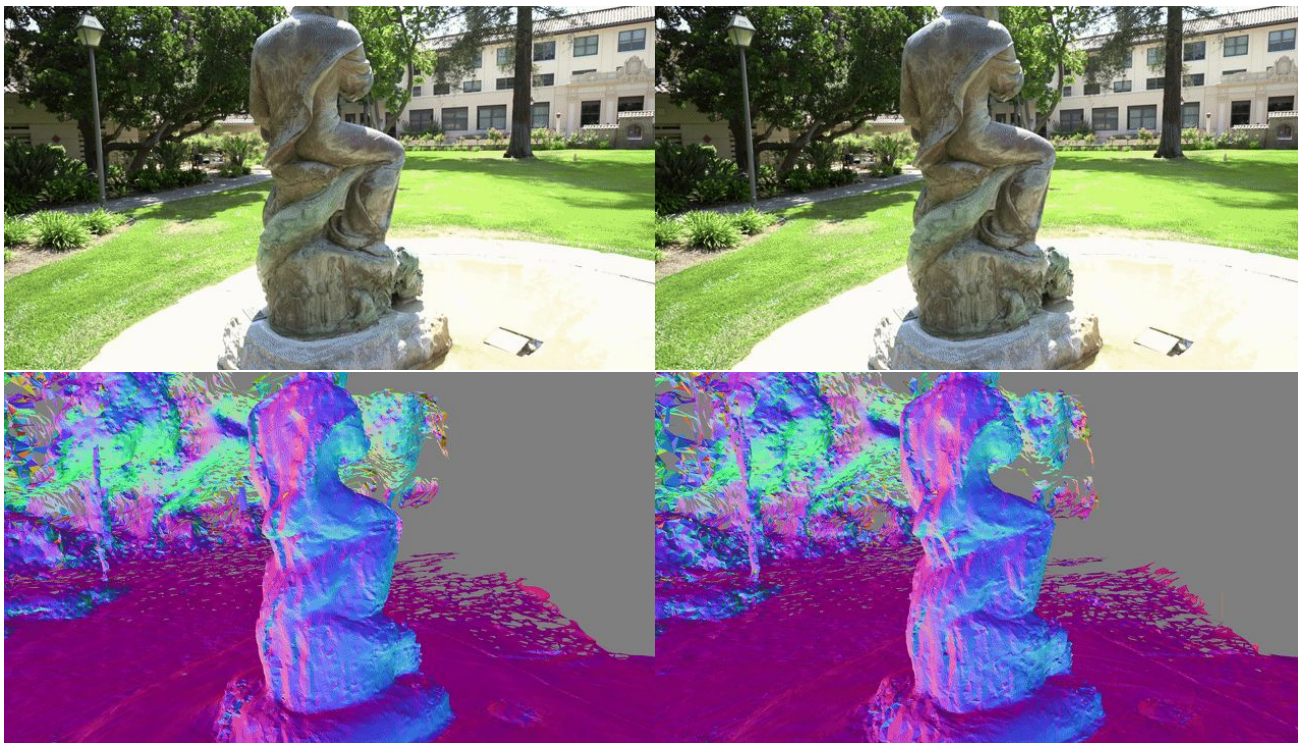
# Finding 1:

## Encoder Trained with Photometric Loss Excels at Depth Accuracy



- Refinement module also does not significantly improve depth accuracy

# Finding 3: Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch

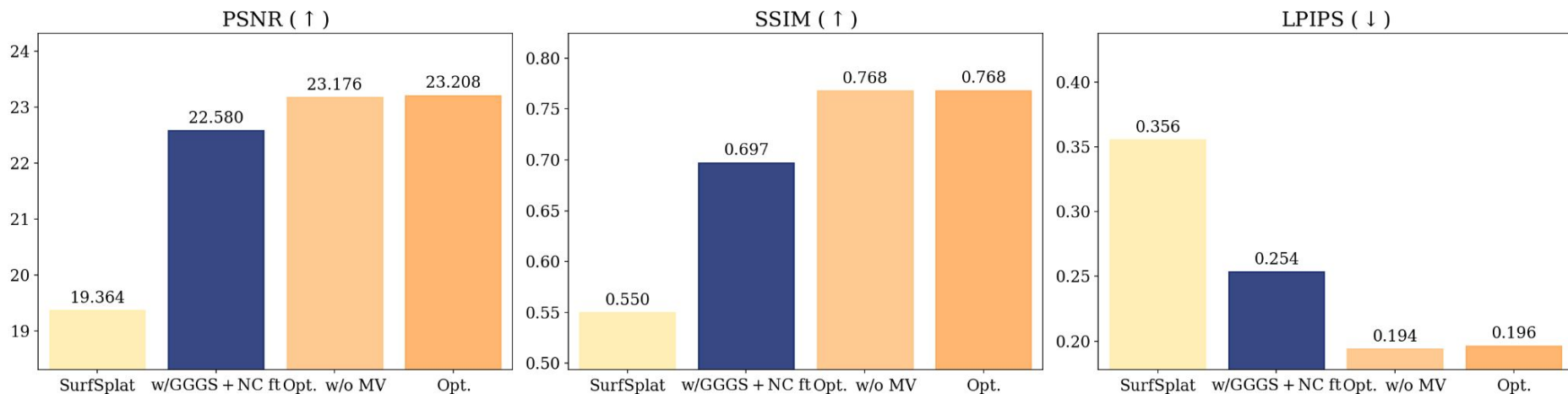


# Finding 3: Fine-Tuning on Normal Consistency Loss Outperforms Training from Scratch



## Finding 5: Fine-Tuned ReSplat Encoder Outperforms DA3, SurfSplat, and Performs on-par with per-scene Optimization

NVS Metrics – Target



# Abstract

*Existing feed-forward Gaussian splatting models focus on visual quality, while the few approaches targeting geometric quality are limited to indoor scenes and image pairs. This thesis focuses on enabling feed-forward models to achieve geometrically accurate 3D representations of unbounded scenes from 8 or 16 sparse views at 256 by 448 resolution, evaluated via extracted meshes and rendered depth maps. Leveraging the recent ReSplat model, we make three major contributions towards this goal:*

*First, we stabilize training by applying a 3D filter to the predicted Gaussians, preventing them from exceeding the Nyquist frequency. Second, we explore the behavior of common self-supervised losses for geometric accuracy in feed-forward models. We find that edge-aware depth smoothness and image warp losses lead to unstable training. Instead, we use a normal consistency loss.*

*Third, we successfully fine-tune a ReSplat encoder, trained with gsplat as its decoder, using the Gaussian Opacity Fields and the Geometry-Grounded Gaussian Splatting (GGGS) decoders alongside the normal consistency loss. These fine-tuned encoders achieve similar geometric accuracy and higher visual quality compared to training from scratch.*

*We evaluate our GGGS fine-tuned encoder and observe strong performance across three key metrics. For mesh reconstruction on the Tanks and Temples dataset, our method improves precision by 41% over the photometric-only encoder, performs on par with per-scene optimization, and surpasses Depth Anything 3 by a factor of three. For depth prediction across MipNeRF360, Tanks and Temples, and DL3DV, our method reduces absolute relative distance by 30% against the photometric-only baseline, 13% against per-scene optimization, and 39% against Depth Anything 3. These improvements come at a minimal cost to visual fidelity, maintaining a PSNR within 0.2dB of the photometric-only encoder.*